

The maximum value of the blue component is '100.' So B_THRESHOLD can be taken as '100.'

Syntax of the functions specified can be had from the Matlab Help. If the image file is not in the current directory, specify the whole path of the file. A semicolon is given after every statement to suppress the output at the command window.

Motion control of the robot

The computer processes the image of the red ball and sends out five different data through its serial COM port. The data depends on the location of the red ball, viz, upper, lower, left, right and centre of the frame, as captured by the camera.

Place the PC-based (USB-based) camera in front of the robot, so that it acts as a visual sensor.

You have to design a robot that connects to the serial port of your PC.

Program the serial port as follows:

1. `obj = serial('port');` creates a serial-port object associated with the serial port specified by the port. If the port does not exist or if it is in use, you will not be able to connect the serial-port object to the device. The syntax is:

```
>>ser=serial('COM1');
```

2. `obj = serial('port','PropertyName','PropertyValue,...');` creates a serial-port object with the specified property names and property values. If an invalid property name or property value is specified, an error is returned and the serial-port object is not created. The syntax is:

```
>> ser= serial('COM1','BaudRate',  
9600,'DataBits',8);
```

3. `fopen(obj);` before you can perform a read or write operation, 'obj' must be connected to the device with the `fopen` function. The syntax is:

```
>>fopen(ser);
```

4. `fprintf(ser,'data');` sends data to the serial port. The syntax is:

```
>>fprintf(ser,'F');
```

5. `fclose(obj);` disconnects 'obj' from the device. The syntax is:

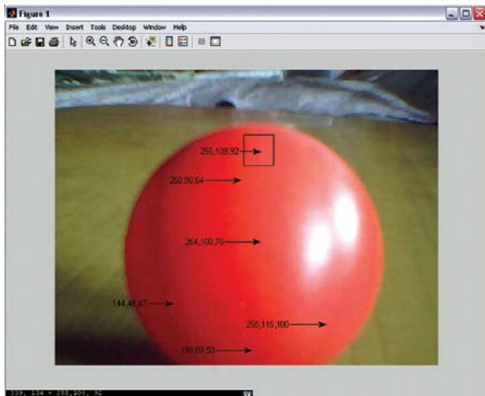


Fig. 5: Screenshot of the red ball

```
>>fclose(ser);
```

Here the robot will be controlled by sending the following codes to the serial port:

1. Key 'F' for forward movement
2. Key 'L' for left movement
3. Key 'R' for right movement
4. Key 'B' for backward movement

Any other code will make the robot stop. The code for implementing motion control of the robot is given at the end of this article.

Testing steps

1. Install Matlab (minimum version: Matlab 7.0.1(R14))

2. Install the camera along with related software

3. Restart your computer

4. Run your camera software and check the image captured by it

5. Take snaps of the red ball from various angles and set the threshold values

6. When all is right, open Matlab software, go to 'm.file' from 'file' menu bar, type the code 'Robotvision.m' (given at the end of the article) and save it. Alternatively, copy the Robotvision.m file from this month's

EFY-CD

7. Click 'run' button in 'Debug' option

8. Hold the red ball in your hand near the camera lens

9. Move the ball left, right, forward and backward. This information will be displayed on your screen. For example, if you move the ball left, you can see 'move left' command on the screen

10. Now, place your camera on top of your robot. The robot will trace the red ball by itself

To check the serial-port data communication, you can use a serial-port monitor, which can be freely downloaded from 'www.download.com/Free-Serial-Port-Monitor/3000-2212_4-70394.html'

Caution. If you want to end execution of your program, go to the Matlab command prompt and press 'Ctrl+C' key. Thereafter, write 'fclose(ser)' against the prompt or else you will not be able to access your serial port further.

EFY note. The source code of this article has been included in this month's EFY-CD.